

# QED-TDDFT: Photon Quantization for Electronic Strong Coupling Modelling

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## 1 Introduction

Strong light-matter coupling has become an increasingly important topic in chemical physics, as it provides a route to generating hybrid molecular states whose properties differ significantly from the uncoupled molecule. In conventional weak-field molecular spectroscopy, the applied electromagnetic field is usually treated as a classical perturbation. The field probes molecular transitions, and the molecular electronic eigenstates are approximated to remain energetically invariant upon the introduction of field interaction, with the radiation only facilitating oscillatory transitions between the states. In this regime, linear-response methods such as TDDFT are sufficient in computing excitation energies, oscillator strengths, and transition dipole moments [1, 2].

However, TDDFT fails in accurately modelling the effects of electronic strong coupling between molecular excitons and photons. Such systems are experimentally studied by placing the desired molecule, embedded in a polymer matrix such as PMMA, in a Fabry-Perot cavity, where the primary mode of the cavity is tuned to be resonant with the primary excitation energy of the molecule being probed. The enhanced electromagnetic field associated with cavity confinement, together with a molecular transition with large transition dipole moment, can lead to light-matter coupling that cannot be treated perturbatively. When the coherent coupling between the molecular transition and the cavity mode exceeds relevant loss and dephasing rates, the stationary eigenstates are no longer purely excitonic or purely photonic. Instead, the molecular excitation and the cavity photon hybridize into lower and upper polariton states. In a minimal one-excitation picture, the relevant basis states are

$$|e, 0\rangle, \quad |g, 1\rangle, \tag{1}$$

where  $|e, 0\rangle$  denotes an electronically excited molecule with zero cavity photons and  $|g, 1\rangle$  denotes the electronic ground state with one photon. Upon strong coupling, the two states form new adiabatic superpositions known as the lower and upper polariton states, with energies lower and higher than the original excitonic energy, respectively.

Unlike molecular spectroscopy, since the energy states of the cavity modes hybridize with those of the molecule, the electromagnetic radiation cannot be dealt with classically anymore. Instead, the relevant state space is the tensor product of the molecular and cavity state spaces. As a result, standard TDDFT is unable to model the hybridization of the photon and electron modes, the amount of photon character a polariton state contains, and how the hybridization is affected by factors such as cavity detuning, light-matter coupling strength, and alignment between the cavity mode polarization and the molecular transition dipole moment.

Quantum-electrodynamical density-functional (QED-TDDFT), as developed by Ruggenthaler *et al.* and expanded upon significantly by Yang *et al.*, provides a new, rigorous but computationally-efficient method of modelling electronic strong coupling as a linear response problem.[3, 4]. In QED-TDDFT,

the cavity modes are quantized into a photonic Fock space, and the solutions incorporate both molecular excitation and deexcitation amplitudes as in standard TDDFT, as well as cavity excitation and deexcitation amplitudes, from which the electronic and photonic character of each polaritonic state can be extracted after proper normalization. Yang *et al.* demonstrated that QED-TDDFT could be used to perform basic computational modelling of the polaritonic branching of small molecules, such as formaldehyde and ethene, in a Fabry-Perot cavity. QED-TDDFT was used to investigate the effects of neglecting deexcitation amplitudes and other energetic terms, as well as computationally tuning facets of the cavity setup, such as primary cavity mode frequency, coupling strength, and cavity mode polarization, on polaritonic branching.[4] Moreover, QED-TDDFT was shown to provide far better qualitative agreement for molecules with energetically close excited states, which interact with each other to create deviations in the strong coupling regime from the predictions of simple two-state models.

The present project applies this framework to larger fluorophores relevant to optical spectroscopy and biophysical labeling: Coumarin 343 and NBD-amine. While these dyes are practically more useful to model due to their prevalence in polaritonic studies, they introduce major electronic-structure challenges. Their low-lying optical states involve long-range electron correlations from delocalization, charge-transfer-like contributions, extreme conformational sensitivity, and strong environmental dependence. Consequently, this work is designed not only to ask whether QED-TDDFT produces polaritonic splitting, but also whether it does so reliably enough for qualitative and quantitative interpretation of realistic dye spectra.

In this paper, I first assess qualitative cavity trends: increasing coupling strength, rotating the cavity mode polarization, and scanning cavity frequency through resonance. I then extend this to generate a 2-D cavity frequency and coupling strength map of the polaritonic energies. Second, I compare QED-TDDFT model approximations for NBD-amine in the same vein as that done for formaldehyde by Yang *et al.*, and highlight differences between NBD-amine and formaldehyde.[4]. Finally, I discuss why quantitative agreement for Coumarin 343 and NBD-like dyes remains difficult, especially in PMMA-like environments modeled by continuum solvation, with respect to molecular properties and computational parameters, especially basis set and functional choice.

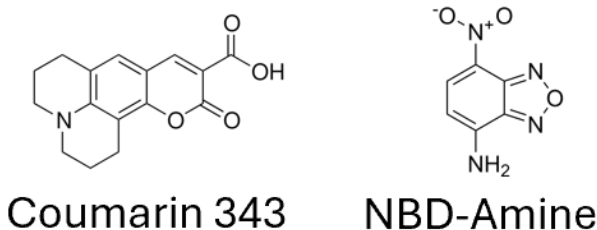


Figure 1: Structural formulae of Coumarin 343 and NBD-amine.

## 2 Methods

### 2.1 Weak-field TDDFT

For an  $N$ -electron molecule in a classical electromagnetic field, the Born–Oppenheimer electronic Hamiltonian can be written in atomic units as

$$\hat{H}(t) = \sum_{i=1}^N \left[ \frac{1}{2} (\hat{\mathbf{p}}_i + \mathbf{A}(\hat{\mathbf{r}}_i, t))^2 - \phi(\hat{\mathbf{r}}_i, t) - \sum_A \frac{Z_A}{|\hat{\mathbf{r}}_i - \mathbf{R}_A|} \right] + \sum_{i < j}^N \frac{1}{|\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j|} + E_{\text{nn}}, \quad (2)$$

The electromagnetic fields are related to the scalar and vector potentials by

$$\mathbf{E}(\mathbf{r}, t) = -\nabla\phi(\mathbf{r}, t) - \frac{\partial\mathbf{A}(\mathbf{r}, t)}{\partial t}, \quad \mathbf{B}(\mathbf{r}, t) = \nabla \times \mathbf{A}(\mathbf{r}, t). \quad (3)$$

Expanding the momentum term gives

$$\frac{1}{2}(\hat{\mathbf{p}}_i + \mathbf{A})^2 = \frac{\hat{\mathbf{p}}_i^2}{2} + \frac{1}{2}[\hat{\mathbf{p}}_i \cdot \mathbf{A} + \mathbf{A} \cdot \hat{\mathbf{p}}_i] + \frac{1}{2}\mathbf{A}^2. \quad (4)$$

In the weak-field approximation,  $\mathbf{A}$  and  $\phi$  are classical fields. The term linear in  $\mathbf{A}$  gives the leading one-photon first-order interaction. Transforming to the electric-dipole length gauge, the weak-field Hamiltonian is written as

$$\hat{H}(t) = \sum_{i=1}^N \left[ -\frac{1}{2}\nabla_i^2 - \sum_A \frac{Z_A}{|\hat{\mathbf{r}}_i - \mathbf{R}_A|} \right] + \sum_{i<j}^N \frac{1}{|\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j|} + E_{\text{nn}} - \hat{\boldsymbol{\mu}} \cdot \mathbf{E}(t). \quad (5)$$

where

$$\hat{\boldsymbol{\mu}} = -\sum_{i=1}^N \hat{\mathbf{r}}_i + \sum_A Z_A \mathbf{R}_A \quad (6)$$

is the molecular dipole operator.

At fixed nuclear geometry, the nuclear contribution  $-\sum_A Z_A \mathbf{R}_A \cdot \mathbf{E}(t)$  is a time-dependent scalar shift and does not affect the electronic excitation structure, and is therefore safely neglected.

TDDFT replaces the exact interacting electronic problem by a Kohn–Sham reference and evaluates the first-order response of the density to a weak external perturbation. In the Casida formulation, the response is expanded in the occupied–virtual Kohn–Sham transition space. The corresponding excitation operator is

$$\hat{Q}_I^\dagger = \sum_{ai} \left[ X_{ai}^I \hat{a}_a^\dagger \hat{a}_i - Y_{ai}^I \hat{a}_i^\dagger \hat{a}_a \right], \quad (7)$$

where  $i$  labels occupied Kohn–Sham orbitals,  $a$  labels virtual Kohn–Sham orbitals, and  $X$  and  $Y$  are the forward and backward components of the density response. Projecting the linear-response equations onto this particle–hole space gives the Casida equation,

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{A} \end{pmatrix} \begin{pmatrix} \mathbf{X}_I \\ \mathbf{Y}_I \end{pmatrix} = \Omega_I \begin{pmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{0} & -\mathbf{1} \end{pmatrix} \begin{pmatrix} \mathbf{X}_I \\ \mathbf{Y}_I \end{pmatrix}. \quad (8)$$

where

$$A_{ai,bj} = (\epsilon_a - \epsilon_i)\delta_{ab}\delta_{ij} + \langle ai | f_{\text{Hxc}} | jb \rangle, \quad B_{ai,bj} = \langle ai | f_{\text{Hxc}} | bj \rangle. \quad (9)$$

Here  $f_{\text{Hxc}}(\mathbf{r}, \mathbf{r}') = |\mathbf{r} - \mathbf{r}'|^{-1} + f_{\text{xc}}(\mathbf{r}, \mathbf{r}')$  is the Hartree–exchange–correlation kernel, which describes the self-consistent change in the Kohn–Sham potential induced by a density perturbation. The Tamm–Dancoff approximation ignores de-excitations, giving the simple eigenvalue problem

$$\mathbf{A}\mathbf{X}_I = \Omega_I \mathbf{X}_I. \quad (10)$$

Conventional TDDFT and TDA compute molecular excitation energies in response to a classical weak field. They do not include the photon field as a quantum coordinate and do not directly produce polaritonic eigenstates.

## 2.2 Quantized cavity field and Pauli–Fierz Hamiltonian

In the strong-coupling regime, the relevant cavity mode must be included as a quantized degree of freedom. The vector potential is expanded in cavity modes as

$$\hat{\mathbf{A}}(\mathbf{r}) = \sum_{\alpha} \sqrt{\frac{1}{\omega_{\alpha} V_{tot}}} \sin\left(\frac{\alpha\pi z}{L_z}\right) \boldsymbol{\epsilon}_{\alpha} \left(\hat{b}_{\alpha} + \hat{b}_{\alpha}^{\dagger}\right), \quad (11)$$

where  $\hat{b}_{\alpha}^{\dagger}$  and  $\hat{b}_{\alpha}$  create and annihilate photons in mode  $\alpha$  with frequency  $\omega_{\alpha}$ .  $V_{tot} = L_x L_y L_z$  is the volume of the cavity, with the mode propagating on the z-axis, and  $\boldsymbol{\epsilon}_{\alpha}$  is the polarization of the photon. Equivalently, each photon mode can be represented by a displacement coordinate and conjugate momentum,

$$\hat{q}_{\alpha} = \frac{1}{\sqrt{2\omega_{\alpha}}} \left(\hat{b}_{\alpha} + \hat{b}_{\alpha}^{\dagger}\right), \quad \hat{p}_{\alpha} = -i\sqrt{\frac{\omega_{\alpha}}{2}} \left(\hat{b}_{\alpha} - \hat{b}_{\alpha}^{\dagger}\right). \quad (12)$$

The cavity-mode photon Hamiltonian is

$$\hat{H}_{ph} = \sum_{\alpha} \left[ \frac{1}{2} \hat{p}_{\alpha}^2 + \frac{1}{2} \omega_{\alpha}^2 \hat{q}_{\alpha}^2 \right]. \quad (13)$$

The optical wavelength is much larger than the molecular length scale, so the field is approximately constant over the molecule. Thus, the long-wavelength approximation is used, and the molecule couples to the total molecular dipole rather than to a spatially varying field. Retaining only the cavity mode near-resonant with the transition of interest, the momentum-gauge Hamiltonian may be written as

$$\hat{H} = \sum_{i=1}^N \left[ \frac{1}{2} \left( \hat{\mathbf{p}}_i + \hat{\mathbf{A}}(\mathbf{r}_i) \right)^2 - \sum_A \frac{Z_A}{|\hat{\mathbf{r}}_i - \mathbf{R}_A|} \right] + \sum_{i<j}^N \frac{1}{|\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j|} + E_{nn} + \hat{H}_{ph}, \quad (14)$$

The momentum-gauge Hamiltonian is then transformed into the length gauge by the unitary transformation

$$\hat{U} = \exp \left[ -i \sum_{\alpha} \frac{\boldsymbol{\lambda}_{\alpha} \cdot \hat{\boldsymbol{\mu}}}{\omega_{\alpha}} \hat{p}_{\alpha} \right], \quad \hat{U}^{\dagger} \hat{q}_{\alpha} \hat{U} = \hat{q}_{\alpha} - \frac{\boldsymbol{\lambda}_{\alpha} \cdot \hat{\boldsymbol{\mu}}}{\omega_{\alpha}}. \quad (15)$$

Here  $\boldsymbol{\lambda}_{\alpha}$  is the effective cavity coupling vector for mode  $\alpha$ , which is directly proportional to the cavity mode polarization  $\boldsymbol{\epsilon}_{\alpha}$ .

$$\boldsymbol{\lambda}_{\alpha} = \sqrt{\frac{2}{V_{tot}}} \sin\left(\frac{\alpha\pi z_0}{L_z}\right) \boldsymbol{\epsilon}_{\alpha}. \quad (16)$$

This gives the long-wavelength length-gauge Pauli–Fierz Hamiltonian,

$$\begin{aligned} \hat{H} = & \underbrace{\left[ \sum_{i=1}^N \left( -\frac{1}{2} \nabla_i^2 - \sum_A \frac{Z_A}{|\hat{\mathbf{r}}_i - \mathbf{R}_A|} \right) + \sum_{i<j}^N \frac{1}{|\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j|} + E_{nn} \right]}_{\hat{H}_{mol}} \\ & + \underbrace{\sum_{\alpha} \left[ \frac{1}{2} \hat{p}_{\alpha}^2 + \frac{1}{2} \omega_{\alpha}^2 \hat{q}_{\alpha}^2 \right]}_{\hat{H}_{ph}} - \underbrace{\sum_{\alpha} \omega_{\alpha} \hat{q}_{\alpha} (\boldsymbol{\lambda}_{\alpha} \cdot \hat{\boldsymbol{\mu}})}_{\hat{H}_{coup}} + \underbrace{\frac{1}{2} \sum_{\alpha} (\boldsymbol{\lambda}_{\alpha} \cdot \hat{\boldsymbol{\mu}})^2}_{\hat{H}_{DSE}}. \end{aligned} \quad (17)$$

The four terms are, respectively, the original molecular Hamiltonian, the photon cavity Hamiltonian, the light–matter coupling, and the dipole self-energy. The dipole self-energy is the quadratic dipole

contribution required by the length-gauge Pauli–Fierz Hamiltonian.

### 2.3 QED-TDDFT response equation

QED-TDDFT applies the same response logic as TDDFT, but to the Pauli–Fierz Hamiltonian. The electronic response operator is expanded to include photon creation and annihilation terms,

$$\hat{Q}_I^\dagger = \sum_{ai} \left[ X_{ai}^I \hat{a}_a^\dagger \hat{a}_i - Y_{ai}^I \hat{a}_i^\dagger \hat{a}_a \right] + \sum_{\alpha} \left[ M_{\alpha}^I \hat{b}_{\alpha}^\dagger - N_{\alpha}^I \hat{b}_{\alpha} \right]. \quad (18)$$

Thus the response vector becomes

$$\mathbf{Z}_I = \begin{pmatrix} \mathbf{X}_I \\ \mathbf{Y}_I \\ \mathbf{M}_I \\ \mathbf{N}_I \end{pmatrix}. \quad (19)$$

The molecular Hamiltonian yields the ordinary TDDFT blocks  $\mathbf{A}$  and  $\mathbf{B}$ . The Pauli–Fierz terms add only three new kinds of blocks: a DSE correction to the electronic response, electron–photon coupling blocks, and a diagonal photon block. Define the cavity-projected transition dipole

$$\lambda_{ai}^{\alpha} = \boldsymbol{\lambda}_{\alpha} \cdot \boldsymbol{\mu}_{ai}, \quad \boldsymbol{\mu}_{ai} = \langle \phi_a | \hat{\boldsymbol{\mu}} | \phi_i \rangle. \quad (20)$$

The DSE contribution is

$$\Delta_{ai,bj} = \sum_{\alpha} \lambda_{ai}^{\alpha} \lambda_{bj}^{\alpha}, \quad (21)$$

and the light–matter coupling gives

$$g_{ai}^{\alpha} = \sqrt{\frac{\omega_{\alpha}}{2}} \lambda_{ai}^{\alpha}. \quad (22)$$

The QED-TDDFT response equation in Casida form is

$$\begin{pmatrix} \mathbf{A} + \boldsymbol{\Delta} & \mathbf{B} + \boldsymbol{\Delta} & \mathbf{g}^\dagger & \tilde{\mathbf{g}}^\dagger \\ \mathbf{B} + \boldsymbol{\Delta} & \mathbf{A} + \boldsymbol{\Delta} & \mathbf{g} & \tilde{\mathbf{g}} \\ \mathbf{g} & \mathbf{g} & \boldsymbol{\omega} & \mathbf{0} \\ \tilde{\mathbf{g}} & \tilde{\mathbf{g}} & \mathbf{0} & \boldsymbol{\omega} \end{pmatrix} \begin{pmatrix} \mathbf{X}_I \\ \mathbf{Y}_I \\ \mathbf{M}_I \\ \mathbf{N}_I \end{pmatrix} = \Omega_I \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} \mathbf{X}_I \\ \mathbf{Y}_I \\ \mathbf{M}_I \\ \mathbf{N}_I \end{pmatrix}. \quad (23)$$

The matrices  $\mathbf{g}$  and  $\tilde{\mathbf{g}}$  are the rotating and counter-rotating electron–photon coupling blocks. The electronic response blocks remain the same size, while a small number of photon amplitudes and low-rank coupling blocks are added. For a single cavity mode, the response dimension changes only from  $2ov$  to  $2ov + 2$ , where  $o$  and  $v$  are the numbers of occupied and virtual orbitals, allowing calculations to be almost as efficient and fast as those of standard TDDFT.

### 2.4 Approximations and model hierarchy

The calculations use the Born–Oppenheimer, long-wavelength, single cavity mode, and linear-response approximations. Linear response remains appropriate in strong coupling because the cavity coupling is included in the Hamiltonian; the weak quantity is the external probe used to interrogate the coupled electron–photon system. It breaks down for strong external driving, nonlinear spectroscopy, multiphoton dynamics, or large nonequilibrium photon populations.

Additional response models are obtained by neglecting selected blocks in Eq. 23, as discussed by Yang *et al.* The eight models considered here are summarized in Table 1.

Table 1: QED-TDDFT approximation hierarchy.

| Electronic response | Model | Y, B | $\Delta$ | $\tilde{g}, N$ |
|---------------------|-------|------|----------|----------------|
| Full TDDFT          | PF    | ✓    | ✓        | ✓              |
|                     | Rabi  | ✓    | ✗        | ✓              |
|                     | RWA   | ✓    | ✓        | ✗              |
|                     | JC    | ✓    | ✗        | ✗              |
| TDA                 | PF    | ✗    | ✓        | ✓              |
|                     | Rabi  | ✗    | ✗        | ✓              |
|                     | RWA   | ✗    | ✓        | ✗              |
|                     | JC    | ✗    | ✗        | ✗              |

TDA neglects electronic de-excitation amplitudes and is most appropriate when the ground state is stable and backward response is small. The rotating-wave approximation neglects counter-rotating terms and is most reliable near resonance when the coupling is small compared with the bare electronic and photon frequencies. Neglecting the DSE is safest at weak or moderate coupling; at stronger coupling it is part of the full Pauli–Fierz description and may shift polariton energies. The TDA-JC model used in most of this work combines all three reductions. For a single cavity mode it becomes

$$\begin{pmatrix} \mathbf{A} & \mathbf{g}^\dagger \\ \mathbf{g} & \omega_\alpha \end{pmatrix} \begin{pmatrix} \mathbf{X}_I \\ M_I \end{pmatrix} = \Omega_I \begin{pmatrix} \mathbf{X}_I \\ M_I \end{pmatrix}. \quad (24)$$

This reduced model is useful for qualitative trends in the weak-to-moderate strong-coupling regime, but it is not equivalent to the full Pauli–Fierz QED-TDDFT response theory.

## 2.5 Computational details

All calculations were carried out using PySCF on the LBNL computing cluster. [14] QED-TDDFT was implemented via the Python library created by Yang *et al.* [4] Unless otherwise indicated, ground-state geometries were optimized at the B3LYP/cc-pVDZ level. Single-point ground-state and excited-state calculations were performed using RKS B3LYP/6-31+G\*\* with ddCOSMO using a dielectric constant of  $\epsilon = 3$  to approximate a PMMA-like environment. Density fitting was employed with aug-cc-pVDZ.

For each molecule, standard TDA-TDDFT calculations were first used to determine the lowest relevant singlet excitation energy. This energy was used as the resonant cavity frequency for TDA-JC QED-TDDFT calculations, and was detuned for cavity frequency scans. Coupling-strength scans varied the cavity coupling parameter  $\lambda$ , polarization scans varied the angle between the transition dipole and cavity polarization vector, and detuning scans varied the cavity frequency  $\omega_\alpha$  around the computed singlet transition energy. The cavity frequency, coupling, and cavity polarization are all parameters included in the QED-TDDFT library. All TDDFT and QED-TDDFT scans for single molecules calculated the first five excitation frequencies, while for the two-molecule case the first ten excitation frequencies were considered.

For NBD-amine, additional coupling-strength scans were performed across multiple QED-TDDFT approximations, including Pauli–Fierz, Rabi, rotating-wave, and Jaynes–Cummings-type models, with and without the Tamm–Dancoff approximation.

## 3 Results and Discussion

### 3.1 Qualitative assays of polarization and coupling strength

The first set of qualitative tests examines whether the computed polaritonic splitting responds correctly to the two most direct cavity parameters: coupling strength and polarization alignment. From the light-matter coupling operator, the electron-photon coupling is proportional to the projection of the transition dipole onto the cavity polarization,

$$g_{ai}^{(\alpha)} \propto \epsilon_{\alpha} \cdot \boldsymbol{\mu}_{ai} = |\boldsymbol{\mu}_{ai}| \cos \theta. \quad (25)$$

Therefore, increasing  $|\boldsymbol{\lambda}|$  should increase the splitting between lower and upper polariton branches, while rotating the transition dipole away from the cavity polarization should suppress the splitting. Scans of these two factors for Coumarin 343 and NBD-amine, with the cavity computationally tuned to be resonant with their ground-to-first-excited-state transition, were conducted as a preliminary test to validate the functionality of the programs. Coupling strength was probed between 0 and 0.01 a.u., with 0.01 a.u. used here as a strong-coupling computational benchmark that produces visibly resolved polaritonic splitting in the model. Cavity frequency was set to the excitonic frequency. In the coupling scan, the polarization was aligned with the transition dipole moment, while in the polarization scan, coupling was set to 0.01 a.u.

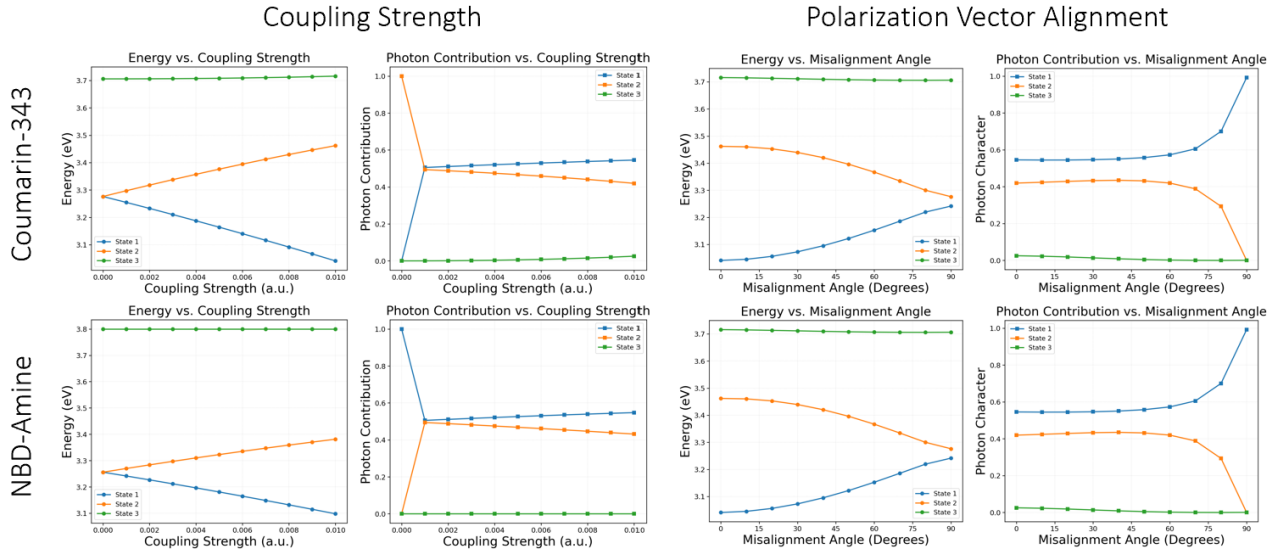


Figure 2: Photon character and polaritonic energy as a function of coupling strength and polarization vector alignment.

As expected, the results qualitatively agree with the theoretical predictions, and resemble the figures also seen with the smaller molecules tested by Yang *et al.*[4] Coumarin 343 displays a higher degree of polaritonic branching of its second excited transition upon an increase in coupling strength, due to a greater magnitude of its transition dipole moment than that of the second transition of NBD-amine. The magnitudes of these transition dipoles can be seen in Table 2, in Section 3.4. Variability in the zero-coupling photon character of the lower and upper polariton states emerges from the fact that the two states are degenerate, and at zero-coupling either one could be considered the exciton or the photon state.

### 3.2 Cavity-frequency detuning: avoided crossings and 2-D frequency-coupling scans

The second qualitative assay is the cavity-frequency detuning scan. Zero coupling produces a crossing between the bare molecular excitation and the bare photon mode at the cavity frequency, which can be thought of as the diabatic representations in the coupling regime. Nonzero coupling lifts this degeneracy and produces an avoided crossing between two new adiabats. In a simple two-state model, this coupling should generate standard avoided crossing curves, but in QED-TDDFT, deviations from these curves may be seen due to mixing of the polaritonic states with higher-energy excitons with non-negligible transition dipole moments of the molecule. As such, cavity-frequency scans were performed for both Coumarin 343 and NBD-amine. In addition, a two-dimensional scan over both cavity frequencies and coupling strength was performed for NBD-amine to visualize the energy surfaces and the concerted effects of detuning and coupling (the latter of which is related to the cavity volume and the number of molecules in the cavity, as will be discussed later) on the degree of polaritonic splitting. Coupling was set to 0.01 a.u in the one-dimensional scan.

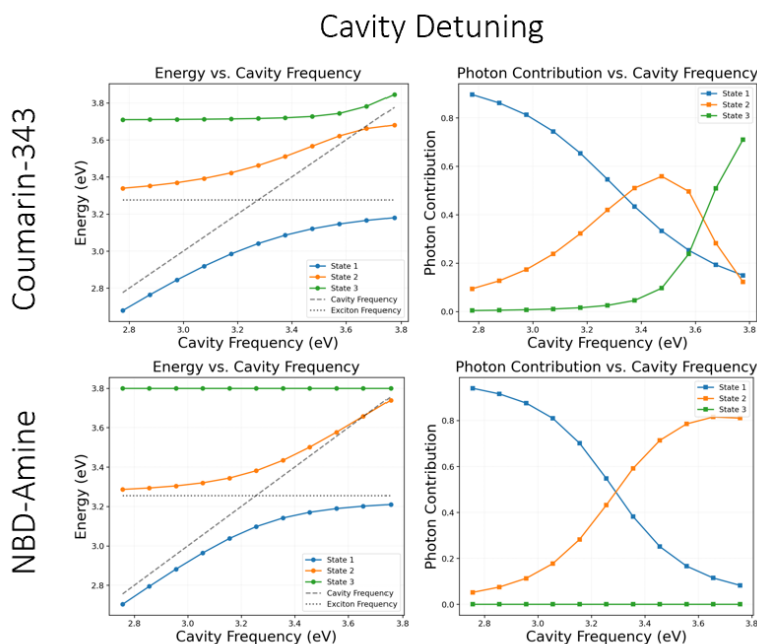


Figure 3: Photon character and polaritonic energy as a function of cavity frequency.

As expected, QED-TDDFT reproduces the avoided crossing diagrams for both Coumarin 343 and NBD-amine. As hinted by the polarization vector and coupling scans, Coumarin 343 shows stronger deviations from a simple two-state polariton picture because its nearby second electronic excitation has a considerably high transition dipole moment, allowing it to mix more appreciably with the polaritonic branches. By contrast, the corresponding low-lying transition of NBD-amine does not, so the avoided crossing remains closer to the two-state model.

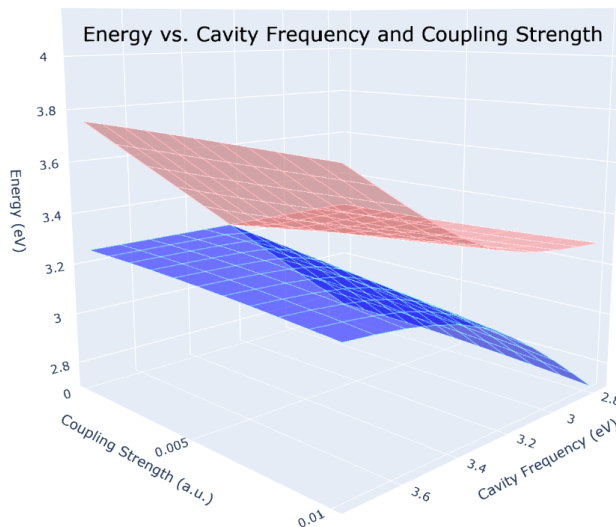


Figure 4: Two-dimensional frequency-coupling strength scan for NBD-amine polaritonic energies.

The two-dimensional scan agrees qualitatively with theoretical reasoning and experimental observation. As expected, as coupling is increased, the crossing of the photon and exciton paths turns into an avoided crossing. Thus, QED-TDDFT is a powerful computational basis to model and visualize the physical effects of polaritonic splitting, even with larger molecules such as NBD-amine.

### 3.3 Comparison of different TDA models via coupling strength scans of NBD-amine

As another check of the functionality of QED-TDDFT for fluorophores such as NBD-amine, the four TDA models were compared to each other: TDA-PF, TDA-RWA, TDA-Rabi, and TDA-JC. TDA-JC was taken as the base model, since it neglected the greatest amount of quantities. Due to issues with the computational cost and convergence time of the other TDDFT methods, this investigation was curtailed to only compare the TDA methods for NBD-amine. Energies and photon contributions as functions of the coupling strength were compared for the primary upper and lower polariton branches for NBD-amine under the four models.

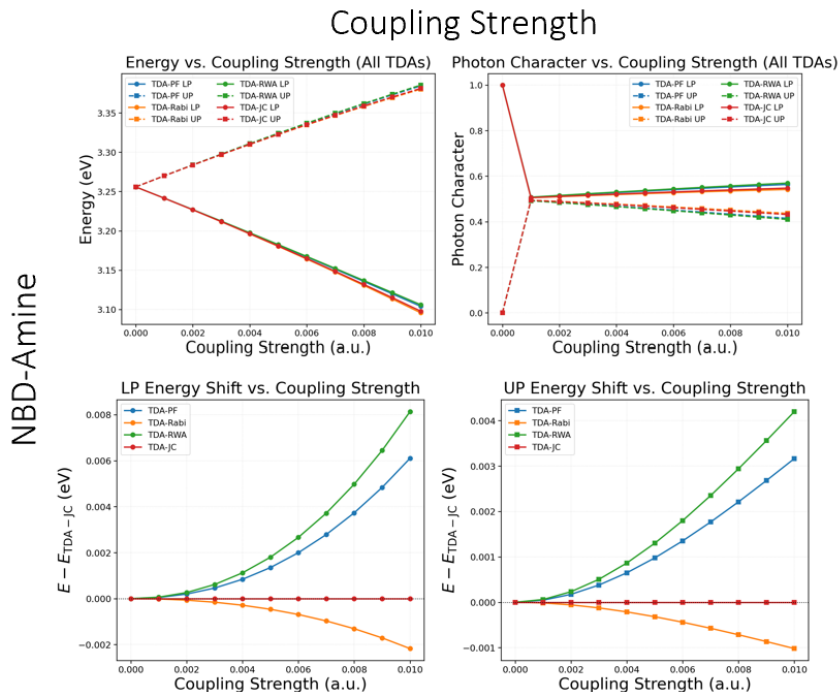


Figure 5: Comparison between TDA models of photon character and polaritonic state energies vs. coupling strength. LP refers to the lower polariton, while UP refers to the upper polariton.

The same qualitative trends seen in formaldehyde were also seen for NBD-amine. The counter-rotating terms included in the Rabi model reduce both the lower and upper polariton energies to a smaller extent ( $\approx -0.001$  eV at 0.01 a.u. coupling), while the dipole self-energy included in the RWA model increases both polaritonic state energies to a larger extent ( $\approx 0.004$  eV at 0.01 a.u. coupling). As discussed by Yang et al., the counter-rotating terms are indeed approximately 1/4th the magnitude of the dipole self-energy, even for NBD-amine. [4] This leads to an overall increase in the polaritonic energies when both are included in the total Pauli-Fierz model. Another observation is that the energy shifts of the lower polariton state seem to be double that of the errors of the upper polariton state. This result is to be investigated further; it is unclear if this is a characteristic of this particular molecule, the molecule initialization methodology used in this project, or of the methods in general. Nevertheless, the range of errors that arise from the various TDA models are lower than the error from different geometry initializations and the single-point DFT calculations (discussed in the next section), and can safely be neglected in these small coupling regimes.

### 3.4 Issues with quantitative modelling of polaritonic splitting

While QED-TDDFT qualitatively reproduces physical phenomena with respect to polaritonic splitting of the larger dyes, it remains significantly less accurate for quantitative modelling of the energetics and photon contributions of the polaritons. The main contribution to these discrepancies between theory and experiment do not come from the linear response step itself, but rather largely from the ground-state geometry optimization and single-point DFT calculations. Large, conjugated molecules such as Coumarin 343 and NBD-amine have significant long-range correlations between electrons and many possible metastable geometries due to relaxations of the delocalized electron system. Moreover, NBD-amine also tends to undergo charge-transfer interactions.

The first major source of error is the geometry optimization. Due to soft torsions and many

metastable geometric states, both Coumarin 343 and NBD-amine (with higher difficulty for Coumarin 343) encounter significant issues during geometry optimization, especially if an incorrect basis is selected. Geometry optimization should also consider these torsions and relative electron density redistributions through the use of basis sets with polarization functions, as well as diffuse functions for NBD-amine, which tends to form charge-transfer states. Such a basis set, 6-31+G(d,p), was used for single-point calculations. For the geometry optimization, cc-pVDZ was used for computational efficiency. Although cc-pVDZ includes polarization functions, it lacks diffuse functions, which may be important for charge-transfer-like states, especially in NBD-amine. Future work should therefore test optimized geometries from larger or more diffuse basis sets, such as aug-cc-pVDZ or 6-31+G(d,p)/6-31++G(d,p), together with conformer searches and frequency checks. Moreover, density fitting was also conducted using aug-cc-pvDZ.

The second major issue is the ground-state single-point KS-DFT calculations, mainly with respect to the functional chosen. For conjugated molecules such as coumarin and NBD-amine, long-range correlation effects are generally poorly modelled by even meta-GGAs, requiring hybrid and double-hybrid functionals to accurately model the ground-state energies and first-excited state transition frequency to within 1 eV. For this paper, B3LYP was chosen since past tests conducted by Lee *et al.* evaluate the MSE of B3LYP as being the most optimal over a variety of functionals, and show B3LYP as having close agreement with EOM-CCSD for Coumarin 343, with around 0.5 eV error in the first excitation frequency. [13] Calculations done in this project for NBD-amine with B3LYP suggested that NBD-amine also had similar errors with B3LYP of approximately 0.7 eV. However, while the errors might not be significant in eV, from a spectroscopic point of view, they represent changes in excitation wavelength of around 100-150 nm, which is enough to even change the behavior of donor-acceptor pairs. Better candidates within the class of hybrid functionals would be range-separated hybrids such as CAM-B3LYP and  $\omega$ -B97X-D, which implement changes in the proportion of DFT exchange and Hartree-Fock exchange based on the distance between electrons to accurately model long-range exchange, and also implement corrections to model long-range dispersion. The most ideal corrections would be to use double-hybrid functionals, although double hybrid-TDDFT is computationally very expensive and not readily available. Future work would consider comparing these results to those obtained from CAM-B3LYP or  $\omega$ -B97X-D.

Table 2: Uncoupled TDA-TDDFT transition properties and comparison to experimental reference energies. Calculated values are from B3LYP/6-31+G(d,p) calculations with PMMA-like ddCOSMO. Experimental energies are obtained by converting excitation wavelengths to energies. Errors are reported only for the  $S_1$  transitions used for experimental comparison.

| Molecule     | State | $E_{\text{calc}} / \text{eV}$ | $ \mu  / \text{a.u.}$ | $f$                  | $E_{\text{exp}} / \text{eV}$ | Error / eV | Percent error |
|--------------|-------|-------------------------------|-----------------------|----------------------|------------------------------|------------|---------------|
| Coumarin 343 | $S_1$ | 3.276                         | 3.182                 | 0.813                | 2.818[16]                    | 0.458      | 16.3%         |
| Coumarin 343 | $S_2$ | 3.706                         | 0.948                 | 0.082                | —                            | —          | —             |
| NBD-amine    | $S_1$ | 3.256                         | 2.137                 | 0.364                | 2.541[17]                    | 0.715      | 28.1%         |
| NBD-amine    | $S_2$ | 3.800                         | $2.3 \times 10^{-4}$  | $4.8 \times 10^{-9}$ | —                            | —          | —             |

Calculated energies represent B3LYP/6-31+G(d,p) calculations with PMMA ddCOSMO, while experimental energies are calculated from wavelengths used to excite Coumarin 343 and NBD. Note that while Coumarin 343 had a peak energy extracted from a spectroscopic scan in PMMA, NBD was tested using a laser close to its peak wavelength, and the experimental value is as such an underestimate of the energy.

The third issue, which amplifies the frequency errors generated from B3LYP, is the solvent model used to simulate the PMMA matrix in the TDDFT calculations. While the ddCOSMO model is a fast model and provides decent modelling of the effect of PMMA, both ddCOSMO and PCM are continuum

models that treat PMMA as a dielectric continuum. However, the large degree of solvatochromicity of NBD-amine implies that even though PMMA is a solid matrix for which the dielectric continuum model would normally be satisfactory, there are still errors in modelling the degree to which PMMA reduces the frequency of the NBD-amine’s first excitation, since the NBD-amine dynamically changes its structure based on the neighboring solvent environment. While nearest-neighbor and other explicit QM/MM solvent models that consider each individual PMMA molecule could be used, the sheer size of the PMMA polymers would require an immense amount of computational effort. These fall under the scope of future explorations of this project to accurately model experimental data.

The last issue is the neglect of broadening of the excitation spectra of the molecules. While the polaritons can be naively modelled as the coupling of cavity modes to pure electronic states, in reality the cavity modes couple to a manifold of vibronic states that would have to be modelled. The primary excited state transition would therefore not also be the vertical transition, causing changes in the uncoupled spectra, and consequently, changing the parameters required to accurately couple the cavity to the molecule. Such issues are only amplified when looking at donor-acceptor interactions, which are discussed further in the next subsection. To accurately model a donor, the geometry of the excited state would have to be optimized, after which its deexcitation energy would have to be calculated using TDDFT. This process is quite inefficient, even after neglecting vibrational effects. Moreover, the Lorentzian lifetime broadening and Gaussian solvent broadening of the spectrum would further skew the results. While Yang *et al.* postulate the modelling of polaritonic spectra using fixed Lorentzian smearing, since inhomogeneous broadening is not as high for benzaldehyde and other small molecules, this same principle does not apply to larger fluorophores such as Coumarin 343 and NBD-amine, which do have significant interactions with the solvent environment. [4] Thus, the extension of this technique to accurately quantitatively model experimental results would require a great amount of memory and temporal resources, and is fraught with computational complications.

Nevertheless, if one were to sufficiently address these issues by optimizing the functional, basis set, and solvent model for their molecule of choice to be studied in a cavity, it would be easier to fit experimental data to models before extrapolating to predictions. The key source of uncertainty would be in the coupling strength, since the true coupling strength comes from the superradiant interaction between a concentrated molecular ensemble, as discussed in the next section.

### 3.5 Future applications of QED-TDDFT

The two most promising avenues for applying and evaluating the effectiveness of QED-TDDFT, once addressing the issues mentioned in the previous subsection, are in accurately quantitatively modelling the polaritonic splitting of molecular ensembles in cavities, and modelling the interactions between donor-acceptor pairs in cavities.

In general, experimental polaritonic systems add a macroscopic concentration of the molecule of interest to a cavity before imaging, causing superradiant effects that trigger coupling between the cavity and the molecules. However, modelling this via QED-TDDFT is essentially impossible, due to the impossibly high cost of incorporating all these molecules into one calculation, which is why the coupling strength is instead manually changed. The major aspect neglected in this is the presence of dark states: when  $N$  molecules couple with a cavity mode, a spectrum of polaritonic states actually form, with the lowest and highest states containing photon character (i.e., the upper and lower polaritons), and  $N - 1$  states in between containing no photon character (i.e., the dark states). It is these dark states that often mediate interactions between polaritonic states in cavity systems, and these are neglected in our calculation. A sample calculation was done with two NBD-amine molecules placed 20 angstrom apart to demonstrate this effect, as well as the difficulties of modelling such dark states.

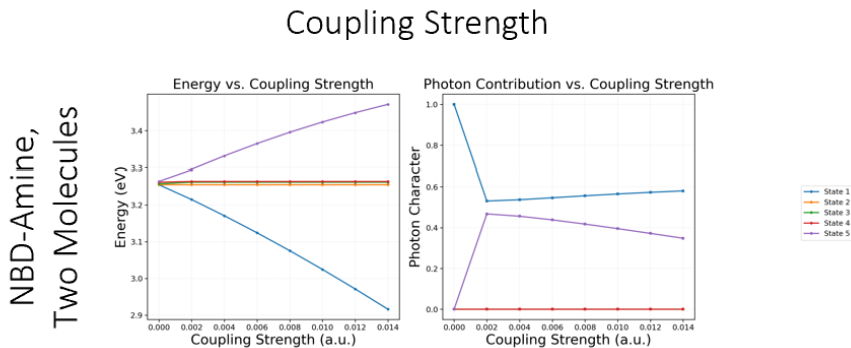


Figure 6: Photonic character and polaritonic state energies as a function of coupling strength for two molecules of NBD-amine placed 20 angstroms apart.

As can be seen in the graphs, while the system qualitatively demonstrates the presence of dark states, it overestimates the number of dark states, as only one dark state would be expected rather than three. This is because even in the uncoupled system, the calculations of the two-molecule system yielded two unusually low-energy second excited frequencies that were not present in the single molecule, but had energies almost exactly the same energy as the first excitation frequency. These were artifacts of unphysical charge-transfer interactions that were modelled by B3LYP for the two-molecule system, which excessively stabilized the second excited energies of the two molecules. This could be alleviated by the use of CAM-B3LYP or other range-separated functionals, but due to temporal constraints, this was not pursued in this current project.

A related extension would be in the modelling of donor-acceptor interactions. Coumarin 343 and NBD-amine, in real solvent environments, in fact interact as a donor and acceptor respectively. In other words, the emission spectrum of Coumarin 343 aligns with the excitation spectrum of NBD-amine, leading to dipole-dipole interaction-mediated transfers in energy upon excitation of Coumarin 343 that lead to the fluorescent emission of NBD-amine instead if they are within a few nanometers of each other in a normal environment. However, in a cavity, the strong coupling to the modes cause the formation of hybrid polaritonic states that couple the excitonic states of both Coumarin 343 and NBD-amine to the cavity, potentially over separations much larger than ordinary near-field donor-acceptor coupling distances. However, as previously mentioned, this is a highly expensive process, as it requires careful optimization of the donor’s excited state and precise modelling of the uncoupled spectral behavior of the donor and acceptor in the PMMA matrix before QED-TDDFT is even attempted. A final extension of this project would be to probe regimes in which the normalized coupling becomes a significant fraction of the bare transition frequency, which is commonly associated with ultrastrong coupling. Ultrastrong coupling arises when numerous low-lying excited states with high dipole moments, generally of a molecular ensemble, interact with the photon cavity modes, leading to strong interexcitonic-photonic interactions.[12] These regimes would seriously strain the linear response method as well, and could possibly necessitate the use of more expensive methods that have been in development, such as QED-CC and real-time extensions. [15]

## 4 Conclusion

QED-TDDFT is a promising, computationally efficient, linear-response technique that expands TDDFT via including photon quantization and considering the linear response of exciton-photon states to light-matter coupling to model polaritonic splitting. It provides an efficient route to qualitative and, for simpler systems, potentially quantitative modelling of polaritonic properties, such as the effects of polaritonic branching with the modification of cavity parameters such as primary cavity mode frequency,

mode polarization, and coupling strength. In this project, the properties of the polaritonic splitting of two commonly-used fluorophores, Coumarin 343 and NBD-amine, were probed using QED-TDDFT. Overall, QED-TDDFT was able to qualitatively model the polaritonic splitting of both molecules. However, it was unable to quantitatively model the splitting with the current methodology, due to inaccuracies in ground-state geometry optimization, ground-state single-point energy calculations, and uncoupled resonance calculations with standard TDDFT, due to long-range exchange-correlation effects, multiple metastable geometries, and strong solvent interactions that were inadequately handled in the current methodology. Moreover, spectral broadening from vibrations, solvent interactions, and state lifetimes were ignored. These led to overestimations of the peak excitation energy by approximately 0.5–0.7 eV, and therefore, could not be satisfactorily extended further with the current choice of functional and basis sets to model more complicated systems, such as the Coumarin 343-NBD-amine donor-acceptor dyad. Future iterations of this project would aim to improve on the methodology, as well as extend it to model the aforementioned dimer and donor-acceptor dyad systems to further evaluate the qualitative accuracy of QED-TDDFT.

## References

- [1] E. Runge and E. K. U. Gross, “Density-Functional Theory for Time-Dependent Systems,” *Phys. Rev. Lett.* **52**, 997–1000 (1984).
- [2] M. E. Casida, “Time-Dependent Density Functional Response Theory for Molecules,” in *Recent Advances in Density Functional Methods*, edited by D. P. Chong (World Scientific, Singapore, 1995), pp. 155–192.
- [3] M. Ruggenthaler, J. Flick, C. Pellegrini, H. Appel, I. V. Tokatly, and A. Rubio, “Quantum-Electrodynamical Density-Functional Theory: Bridging Quantum Optics and Electronic-Structure Theory,” *Phys. Rev. A* **90**, 012508 (2014).
- [4] J. Yang, Q. Ou, Z. Pei, H. Wang, B. Weng, Z. Shuai, K. Mullen, and Y. Shao, “Quantum-Electrodynamical Time-Dependent Density Functional Theory within Gaussian Atomic Basis,” *J. Chem. Phys.* **155**, 064107 (2021).
- [5] E. T. Jaynes and F. W. Cummings, “Comparison of Quantum and Semiclassical Radiation Theories with Application to the Beam Maser,” *Proc. IEEE* **51**, 89–109 (1963).
- [6] A. Mandal, M. A. D. Taylor, B. M. Weight, E. R. Koessler, X. Li, and P. Huo, “Theoretical Advances in Polariton Chemistry and Molecular Cavity Quantum Electrodynamics,” *Chem. Rev.* **123**, 9786–9879 (2023).
- [7] T. E. Li, B. Cui, J. E. Subotnik, and A. Nitzan, “Molecular Polaritonics: Chemical Dynamics Under Strong Light-Matter Coupling,” *Annu. Rev. Phys. Chem.* **73**, 43–71 (2022).
- [8] A. Dreuw and M. Head-Gordon, “Failure of Time-Dependent Density Functional Theory for Long-Range Charge-Transfer Excited States: The Zincbacteriochlorin-Bacteriochlorin and Bacteriochlorophyll-Spheroidene Complexes,” *J. Am. Chem. Soc.* **126**, 4007–4016 (2004).
- [9] T. Yanai, D. P. Tew, and N. C. Handy, “A New Hybrid Exchange-Correlation Functional Using the Coulomb-Attenuating Method (CAM-B3LYP),” *Chem. Phys. Lett.* **393**, 51–57 (2004).
- [10] J. Tomasi, B. Mennucci, and R. Cammi, “Quantum Mechanical Continuum Solvation Models,” *Chem. Rev.* **105**, 2999–3094 (2005).

- [11] F. Lipparini, B. Stamm, E. Cancès, Y. Maday, and B. Mennucci, “Fast Domain Decomposition Algorithm for Continuum Solvation Models: Energy and First Derivatives,” *J. Chem. Theory Comput.* **9**, 3637–3648 (2013).
- [12] R. C. Couto and M. Kowalewski, “Suppressing Non-Radiative Decay of Photochromic Organic Molecular Systems in the Strong Coupling Regime,” *Phys. Chem. Chem. Phys.* **24**, 19199–19208 (2022).
- [13] D. Lee, L. Greenman, M. Sarovar, and K. B. Whaley, “Ab Initio Calculation of Molecular Aggregation Effects: A Coumarin-343 Case Study,” *J. Phys. Chem. A* **117**, 11072–11085 (2013).
- [14] Q. Sun *et al.*, “Recent Developments in the PySCF Program Package,” *J. Chem. Phys.* **153**, 024109 (2020).
- [15] T. S. Haugland, E. Ronca, E. F. Kjørstad, A. Rubio, and H. Koch, “Coupled Cluster Theory for Molecular Polaritons: Changing Ground and Excited States,” *Phys. Rev. X* **10**, 041043 (2020).
- [16] N. Ichiji, Y. Otake, and A. Kubo, “Femtosecond Imaging of Spatial Deformation of Surface Plasmon Polariton Wave Packet during Resonant Interaction with Nanocavity,” *Nanophotonics* **11**, 1321–1333 (2022).
- [17] S. J. Lord, H.-l. D. Lee, R. Samuel, R. Weber, N. Liu, N. R. Conley, M. A. Thompson, R. J. Twieg, and W. E. Moerner, “Azido Push–Pull Fluorogens Photoactivate to Produce Bright Fluorescent Labels,” *J. Phys. Chem. B* **114**, 14157–14167 (2010).